

REVIEW ARTICLE

Properties Calculations of Silica-Based Glasses by Atomistic Simulations Techniques: A Review

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Silica-based glasses are of great importance in various technological fields ranging from fiber optic wave guides, laser optics for initiating fusion reactions, to containers for radioactive waste and biomedical applications. In the past, several works have been devoted to the modeling of structural properties of such materials, and intuitive relationships with experimental macroscopic properties have been addressed. However, the ability to predict properties such as mechanical strength, elastic constants, chemical durability, and conductivity, which are of particular interest since they dictate whether a specific need or application can be met, is becoming an essential tool for glass manufactures. In this review, the results of classical atomistic simulations of the structure, elastic properties, stress–strain diagrams, fracture mechanisms, and transport properties of silicate glasses are presented.

Introduction

Silica-based glasses are of great importance in various technological fields ranging from fiber optic wave guides, laser optics for initiating fusion reactions, containers for radioactive waste, and biomedical applications. Advanced techniques such as X-ray absorption fine structure (XAFS),^{1,2} magic angle spinning NMR (MAS NMR),^{3–9} and neutron diffraction^{10–14} together with molecular dynamics (MD) simulations^{4,15–23} have been crucial in the elucidation of the structural features of several glass compositions. Beyond this, the ability to predict properties such as mechanical strength, elastic constants, chemical durability, and conductivity,²⁴ which are of particular interest since they dictate whether a specific need or application can be met, is becoming an essential tool for glass manufactures. Atomistic simulations techniques are particularly valuable in this area for two reasons. First, they offer additional insight into the microscopic structure and help the resolution of ambiguities in the experimental data, and second, provided an appropriate realistic interaction model is used, they can give quantitative information on transport phenomena, fracture mechanisms, and other physical properties. In this paper, a review of recent progresses obtained in the field of computer simulations of glass properties is provided. Far to be exhaustive of the whole topic, this review will focus on the simulation of the structure, mechanical, and transport properties of silicate glasses by using classical atomistic simulations. Although the discussion is addressed to a wide audience, a general familiarity with molecular mechanics and dynamics methods is assumed.

Several reviews have been devoted in the past to the application of molecular dynamics simulations to the elucidation of the structure of silicate glasses²⁵ and their surfaces;²⁶ however, the simulation of their mechanical properties has never been reviewed.

After a brief discussion of the basic features of the structure of silicate glasses, some of the interatomic potential model recently used will be described followed by a description of the methods employed to study properties of amorphous materials. Then, their applications to a number of silicate glasses will be discussed. The general theme addressed will be how compositional changes alter the atomic structure, and how the changing structures controls important properties such as transport and mechanical properties. Finally, new perspectives on the simulations of glasses will be provided.

Silicate Glass Structure: General Features

The basic structure of silica, SiO_2 , has long been understood in terms of a three-dimensional network of SiO_4 tetrahedra.²⁷ Spectroscopic and scattering (neutron and X-ray) studies indicate that the geometry of the tetrahedra is almost constant throughout the glass, as it is in the crystalline counterparts.²⁸ This geometry is often characterized as short-range order (SRO), which relates the arrangement of the oxygens coordinated to the cations of the glass network. In silica, every tetrahedron is linked to four other tetrahedra by bridging oxygen (BO), forming the network. The variation in the Si–O–Si bond angle connecting two tetrahedra is considered to be one of the main sources of disorder in the structure.

The addition of alkali or alkaline-earth metal cations breaks the silicate network by replacing Si–BO–Si bonds with Si–NBO, where NBO is a nonbridging oxygen. They are thus called network modifiers in contrast to the silicon ions, which are called network formers. According to the Modified Random Network (MRN) model²⁹ at high concentrations, the modifiers percolate through the bulk of the glass and form primary pathways, or channels, which play a key role in the elastic and transport properties of glasses.

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Dr. Alfonso Pedone was born in Atripalda (AV), June 26, 1980. In July 2004, he received his degree in Chemistry at the University of Modena and Reggio Emilia with full marks and honors (110/110 cum laude). In February 2008, he received his Ph.D. in Chemistry at the University of Modena and Reggio Emilia. From May 2008 to May 2009, his Postdoc position was at the department of Chemistry at the University of Modena and Reggio Emilia. Currently, he has a two year contract as a researcher at Scuola Normale Superiore Pisa granted by Telcom Italia: "Progetto Italia Fellowship at Scuola Normale Superiore". He performs research activity in the field of computer simulations based on classical and quantum-mechanical methods applied to the study of structural, dynamical, thermodynamical, mechanical, and spectroscopic properties of silica-based inorganic materials and biomaterials as well as on the simulation of electronic, vibronic, EPR, IR, and NMR spectra of organic dyes encapsulated and grafted on nanostructured inorganic materials. He spent periods of study to Italian and foreign research institutes; in particular, he has been a visiting scientist at the University of Torino, Alfred University, NY (USA), and University of Napoli. He is member of the Italian Chemical Society, Division of Physical-Chemistry and Inorganic Chemistry (SCI), and of the "Consorzio Interuniversitario Nazionale per la Scienza e Tecnologia dei Materiali", (INSTM), a network of university laboratories for the study of the properties and applications of new materials. He has been awarded with the MODENA ROTARY CLUB prize titled "The best scientific brains" in the field of Chemistry Physics and Mathematics and the "Primo Levi Prize 2006" for the best young Italian researcher in chemistry released by the Italian Chemical Society. He took part to several congresses and workshops as speakers and he is coauthor of 24 articles published on international scientific journals.

Interatomic Potential Models

The atomistic approach to model structure and other properties of materials involves the definition of interatomic potential functions to simulate the forces acting between atoms. The total potential energy (U) is generally broken down into pair, three-body, and higher order terms depending on the chemical-physical features of the system under study.

In many cases, especially for oxides and silicates, it is common to approximate U by including only the two body components. They are assumed to depend on the interion distance $r = |\mathbf{r}_i - \mathbf{r}_j|$ and may be usefully decomposed into a long-range Coulombic term and short-range terms that mimic the repulsion between electron charge clouds, van der Waals attraction, and electronic polarizability according to the Born model.³⁰ When relevant, covalent interactions are modeled by three body terms, the simplest of which is the harmonic bond bending.

In 1976, Woodcock et al.³¹ carried out the first MD simulation on vitreous silica using a purely ionic pair potential. Then, representation of the covalent bonding in SiO_2 was introduced by Feuston and Garofalini^{32,33} and subsequently by Vessal et al.³⁴ and Vashishta et al.³⁵

These were derived empirically to model structural features of glasses and, as a consequence of the inclusion of three body terms for reproducing the O–Si–O and Si–O–Si bond angles, they show essentially two drawbacks. (a) They are not transfer-

able to crystalline silicates where the Si–O–Si bond angle can cover a wide range of values (130–180°), and (b) their performance in calculating mechanical properties are poor, the error being of 2 order of magnitude higher.

To obtain a reliable description both of the structural and mechanical properties of oxides, silicate crystals, and silica-based glasses, we recently developed a new pairwise interatomic potential³⁶ based on a rigid ionic model with partial charges to handle the covalent character of silicate systems.

The potential model adopted in most of the works reviewed here is made up by the following three terms: (a) long-range coulomb potential, (b) short-range Morse function, and (c) repulsive contribution C/r^{12} , necessary to model the interactions at high-temperature and pressure. Therefore, the expression for the resulting potential is given by

$$u_{ij}(r) = \frac{z_i z_j e^2}{r} + D_{ij} [1 - e^{-a_{ij}(r - \bar{r}_{ij})}]^2 - 1 + \frac{C_{ij}}{r^{12}} \quad (1)$$

where z_i , z_j , D_{ij} , a_{ij} , $\bar{r}_{ij}$, and C_{ij} are parameters and the indices i and j refer to the different atomic species.

In this framework, the parameters for the functional form chosen were derived from structural and elastic constants of binary oxides by using the relaxed fitting procedure.³⁷

The O–H parameters for the hydroxyl groups were also included by fitting the structure and the free O–H vibrational frequencies of the (001) surface of trydimite crystal, while the H-bond parameters were fitted on the (100) surface of trydimite calculated from quantum-mechanical periodic calculations.³⁸

These have been designed to be consistent with the hydroxyl-SPC water intermolecular parameters developed by Hassanali et al.³⁹ which opened a new scenario in the study of silica-water interactions. The charges and parameters of the developed force-field are reported in Table S.1-2 of Supporting Information as well as the predicted structural and mechanical properties of oxides, silicate crystals, hydroxides and clay minerals in Table S.3-5 of Supporting Information.

Recently, several partial charges rigid ionic interatomic potentials models with a short-range Buckingham term have also been introduced and successfully used to model the structure and ionic diffusion.^{17,19,40–43}

However, it is worth mentioning that the effect of the diverse an often quite distorted local environment can hardly be reproduced by mean field approaches using fixed charges and rigid ionic force fields. A step further into the prediction of dielectric, optical, and medium range structural and dynamical properties is obtained by the inclusion of ionic polarizability via a shell-model approach⁴⁴ that has been shown to be crucial for the correct reproduction of vibrational modes of silicates³⁸ and phosphates⁴⁵ as well as the description of the Si–O–Si angles and Q^n distributions in phospho-silicate glasses.^{46–48}

In this approach, the total charge Z of oxygen ions is split into a core (of charge $Z-Y$) and a massless shell (of charge Y) that are coupled by a harmonic spring. Besides the damped harmonic interaction with the corresponding core, the oxygen shells interact with each other and with cations through a short-range Buckingham term, whereas Coulombic forces act between all species, which usually bear full formal charges.

Another approach consists in the use of a point ion dipolar polarizability, α , which, in the presence of an electric field, V_e , gives rise to a dipole moment, $\mu = \alpha V_e$, and energy of interaction $U = -1/2 \alpha V_e^2$. This approach has the advantage that it is

readily extended to higher order polarizabilities, such as quadrupolar, and so forth.⁴⁹ It has been successfully applied by Wilson et al.^{50,51} to calculate the infrared spectrum for pure ν -SiO₂ and to the simulation of bulk and surface properties of amorphous SiO₂.^{52,53} The only disadvantage of this approach is the independency of the polarizability from the environment, which implies that is undamped at extreme electric fields and can lead to a polarization catastrophe.

Glass Generation via MD Simulations

Bulk glasses are usually simulated via the melt-quench technique in which a system of atoms is either given a crystalline structure or a random distribution within the central cell with three-dimensional PBC. The system is brought to high temperature in order to generate a simulated melt and then cooled down at room temperature.

It is worth mentioning that different authors may use different cooling cycles.⁵⁴ While the short-range structure seems predominantly governed by the interatomic potential parameters,^{55,56} the cooling procedure was found to slightly affect the medium range structure, specifically in the population of the different ring size distributions and in the number of defects formed.⁵⁷

Experience from recent work has led to the following cooling procedure to be used when the partial charges rigid ionic potential model is employed:^{58,59} the system is heated at 6000 K, equilibrated for 100 ps, and subsequently cooled down continuously from 6000 to 300 K in 1140 ps with a nominal cooling rate of 5 K/ps. The temperature is decreased by 0.01 K every time step using the Berendsen thermostat⁶⁰ with the time constant parameter τ_T for the frictional coefficient set to 0.4 ps. Another 100 ps of equilibration at constant energy and 50 ps of data production is performed at 300 K. Integration of the equation of motion is performed using the Verlet leapfrog algorithm with a time step of 2 fs. Coulomb interactions are calculated by the Ewald summation method⁶¹ with a cutoff of 12 Å and an accuracy of 10^{-4} .

When the shell model is employed, a time step of 0.2 fs is required to control the high frequency motion of the core-shell spring during MD simulations and a nominal cooling rate of 10K/ps is normally used.⁴⁷

It is worth noticing that the short time and length scale of computer simulations requires a cooling rate several orders of magnitude higher than typical experimental rates that is known to result in a glass transition temperature significantly higher than the actual one. Because of the smallness of the system, the rate at which energy dissipates implies that in simulations the quenching rate is inversely proportional to the simulated system size. As a consequence, reported quenching rates are not intrinsically fast and a mole of particles is needed to reach the rate observed experimentally.⁵⁷ In fact, it has been reported that small drops of melted alumina-based lanthanide glasses quench at faster rates than larger drops.⁶²

However, previous simulations showed that despite the different T_g a cooling rate around 10 K/ps yields converged and accurate structural properties⁶³ and is used on most MD studies of melt-derived glasses.

Analysis of Static and Dynamic Properties

This section covers an overview of the quantities used to describe the structure and the dynamics of inorganic glasses.

Pair Distribution Functions. Two of the most useful structural properties that can be extracted from the molecular dynamics simulation are the radial distribution function (RDF) and pair distribution function (PDF). These two functions

measure how atoms organize themselves around one another. Specifically, they are proportional to the probability of finding two atoms separated by a distance $r + \Delta r$. The PDF can be expressed as follows

$$g_{ij}(r) = \frac{\sum_{k=1}^M N_j^k(r, \Delta r)}{M/2N_i\rho_jV(r, \Delta r)} \quad (2)$$

where $N_j^k(r, \Delta r)$ is the number of atoms of the j -th species found at the instant t_k in a spherical shell of radius r and thickness Δr , with the shell centered on an atom of the i -th species; $V(r, \Delta r)$ is the volume of the shell; N_i is the number of atoms of the i -th species; $\rho_j = N_j/V$ is the density of atoms of the j -th species. During a simulation run, $N_j^k(r, \Delta r)$ is measured M times and then averaged. By integrating $g_{ij}(r)$ between r_1 and r_2 the coordination number (CN) of the corresponding coordination shells is obtained

$$CN_i(r_1, r_2) = \sum_j \int_{r_1}^{r_2} 4\pi\rho_j r^2 g_{ij}(r) dr \quad (3)$$

By accumulating the PDFs for all the pairs in the system, the RDF and the total correlation function, $T(r)$ can be obtained. The relations between $g_{ij}(r)$, RDF(r), and $T(r)$ can be expressed as follows

$$\text{RDF}(r) = 4\pi r^2 \sum_{ij} g_{ij}(r) \quad (4)$$

$$T(r) = 4\pi r \sum_{ij} g_{ij}(r) \quad (5)$$

The principal advantage of RDF(r) and $T(r)$ is their comparability with X-ray and neutron diffraction data. Nevertheless, to make a quantitative comparison, factors such as scattering length (for neutron diffraction) and instrumental and thermal broadening have to be included.⁶⁴

According to the standard definitions,⁶⁵ the partial structure factors are calculated as the FT of the corresponding partial pair distribution function $g_{ij}(r)$

$$A_{ij}(q) = 1 + 4\pi\rho_0 \int_0^R r^2 [g_{ij}(r) - 1] \frac{\sin(qr)}{qr} \frac{\sin(\pi r/R)}{\pi r/R} dr \quad (6)$$

where ρ_0 is the average atom number density. The window function $\sin(\pi r/R)/(\pi r/R)$ is included to reduce the termination effects resulting from the finite upper limit in the FT of $g_{ij}(r)$.⁶⁶ The cut off length R is usually chosen to be equal to half of the simulation box edge. However, since changing the cut off of the FT also changes details of the transformed data a large box size must be used for simulating structural factors.

The normalized total-scattering structure factor $S(q)$ is obtained as the weighted sum of the partial structure factors, the weights being the coherent neutron scattering lengths $\bar{b}_i$

$$S(q) = \frac{\sum_{ij} c_i c_j \bar{b}_i \bar{b}_j A_{ij}(q)}{(\sum_i c_i \bar{b}_i)^2} \quad (7)$$

where $c_i = N_i/N$ is the relative concentration of the i -th species.

Q_n , Ring Size, and Void Size Distributions. The number n of bridging oxygens associated with the $[\text{SiO}_4]$ tetrahedron is an important index in the description of glass structure. Q_n is defined as a species with n BO associated with it. For example, a Q_3 species of silicon is a Si with 3 BO and one NBO in its tetrahedron. The Q_n distribution represents the percentage of tetrahedra with n BO. Experimentally, the Q_n distribution can be determined by NMR and Raman techniques. MD simulations give the possibility to determine the composition of Q_n distributions, by computing the quantities Q_n^{ijkm} where the upper suffixes indicate the type of network formers bonded to the central species as well as the partitioning of modifier ions around different Q_n species useful, for example, for explaining solubility data and chemical durability.⁴⁶

Another important quantity is the ring size distribution (RSD) that is usually defined as the number of tetrahedra involved in the smallest closed chains formed by BO and network formers in the glass network. Rings that can be decomposed into two or more smaller rings are not counted in the distribution. Although the RSD is not experimentally measurable for silicate glasses at present, most of the structural models in the literature are frequently discussed in terms of it.^{16,67}

The void size distribution (VSD) is the distribution of the occurrence of voids (defined as holes) of radius r and it is important not only for a description of the midrange structure of glasses but also for many diffusion related properties and the behavior of properties at the glass transition temperature. Specifically, the VSD can be indirectly derived from gas diffusion measurements. Therefore it provides an effective way to compare the Intermediate Range Order (IRO) between the experimentally melted glasses and computer simulated glasses. An algorithm based on the geometric Voronoi analysis has been recently developed⁶⁸ and its applications are discussed in the literature.^{68–72} The Voronoi analysis consists of the partitioning of the space in polyhedra, each containing the volume nearest to a given atom. The vertices of these polyhedra are equidistant to at least four atoms in the network, so that it is possible to define spheres centered at the vertices which correspond to the interstitial voids.

The free volume (FV) of different glasses is analyzed in terms of the Delaunay description of the void space into a disordered system.⁶⁸ A set of four atoms contiguous to each other forms a tetrahedron which is called Delaunay tetrahedron (DT). The whole set of the DTs constitutes the Delaunay tessellation which is dual to the Voronoi tessellation, since they fill the space without gaps and without overlaps. The FV is calculated as the summation of all the void spaces inside the DTs that cover the entire box of the simulation.⁶⁸ Therefore, the fractional free volume (FFV) is defined as the fraction of FV over the total volume of the box. This analysis requires that atoms are considered as spheres.

Elastic Properties Calculation. Three basic methods are available for calculating mechanical or elastic properties from static calculations, second derivative, constant stress minimization, and constant strain minimization. These techniques can all be used to obtain the stiffness matrix as well as its inverse, called the compliance matrix. The second derivative method uses a single-point energy calculation to obtain the stiffness matrix elements for a crystalline system, which are defined by

$$C_{ij} = \frac{1}{V} \left(\frac{\partial^2 U}{\partial \varepsilon_i \partial \varepsilon_j} \right) \quad (8)$$

Once the stiffness matrix $\mathbf{C}$ is obtained, several related mechanical properties of anisotropic materials can be derived from their matrix elements or from the matrix elements of the compliance matrix $\mathbf{S}$

$$\mathbf{S} = \mathbf{C}^{-1} \quad (9)$$

The bulk modulus B in the Reuss and Voigt notations is given by

$$\begin{aligned} B_{\text{Reuss}} &= (S_{11} + S_{22} + S_{33} + 2(S_{31} + S_{21} + S_{32}))^{-1} \\ B_{\text{Voigt}} &= \frac{1}{9}(C_{11} + C_{22} + C_{33} + 2(C_{12} + C_{13} + C_{23})) \end{aligned} \quad (10)$$

while the Shear modulus is given by

$$\begin{aligned} G_{\text{Reuss}} &= \frac{15}{4(S_{11} + S_{22} + S_{33} - S_{12} - S_{13} - S_{23}) + 3(S_{44} + S_{55} + S_{66})} \\ G_{\text{Voigt}} &= \frac{1}{15}(C_{11} + C_{22} + C_{33} + 3(C_{44} + C_{55} + C_{66}) - C_{12} - C_{13} - C_{23}) \end{aligned} \quad (11)$$

Poisson's ratios are calculated from the compliance matrix for six components

$$\nu_{xy} = -\frac{S_{21}}{S_{11}}, \quad \nu_{xz} = -\frac{S_{31}}{S_{11}}, \dots, \quad \nu_{zy} = -\frac{S_{32}}{S_{33}} \quad (12)$$

whereas the Young's modulus values along three principal directions are given by

$$E_k^{-1} = S_{kk} \quad (k = 1, 3) \quad (13)$$

The acoustic velocities in a solid can be derived from the bulk and shear moduli of the material, as well as the mass density, d . There are two values, that for a transverse wave, V_T , and that for a longitudinal wave, V_L , which are given by

$$V_T = \sqrt{\frac{G}{d}} \quad \text{and} \quad V_L = \sqrt{\frac{4G + 3B}{3d}} \quad (14)$$

For isotropic materials the stiffness and compliance matrices have 3 independent elements being $C_{11} = C_{22} = C_{33}$, $C_{12} = C_{13} = C_{23}$, $C_{44} = C_{55} = C_{66} = 1/2(C_{11} - C_{12})$ and zero the remaining constants. The Reuss and Voigt formulas are then equivalent and E , G , B and ν satisfy two relations

$$G = \frac{E}{2(1 + \nu)} \quad \text{and} \quad B = \frac{E}{3(1 - 2\nu)} \quad (15)$$

Therefore, only two independent constants are needed to specify the elastic behavior of an isotropic material.

In the framework of the second derivative method implemented in the GULP code,⁷³ the elastic properties are computed for the glasses obtained from the MD simulation runs by using

the following procedure: a Newton–Raphson⁷⁴ energy minimization is performed at constant pressure using the approach of Fletcher and Powell⁷⁵ to update the Hessian matrix. The coulomb term is evaluated by means of the Ewald method in which the real cutoff is determined according to an accuracy of 10^{-8} .⁷⁶ To handle the amorphous character of the glass, a cubic cell with no symmetry (space group $P1$) is used.

Simulation of the Stress–Strain Diagrams and Fracture Mechanisms. Several papers have been devoted to the study of the fracture mechanism of glasses.^{77–81} A method for simulating the stress–strain diagrams and fracture mechanisms recently published by Pedone et al.⁸² is in the following summarized as an example: after molecular dynamics equilibration at constant pressure ($P = 0$ kbar) and temperature using the Berendsen algorithm,⁶⁰ the samples is subjected to uniaxial tensile loading through gradual elongation to determine stress–strain curves. Sample deformation is administered via a stepwise displacement of atoms, $\mathbf{r} = (1 + \epsilon)\mathbf{r}_0$, where $\mathbf{r}$ and $\mathbf{r}_0$ are the displaced and previous positions of an atom at each time step, and ϵ is the applied strain.

Component of stresses are calculated by:

$$\sigma_{\alpha\beta} = -\frac{1}{V} \left(\sum_i p_{i\alpha} p_{i\beta} / m_i + \sum_i r_{i\alpha} f_{i\beta} \right) \quad (16)$$

where $p_{i\alpha}$ and $f_{i\alpha}$ are the α component of the momenta and forces acting on the i -th particle and m_i is its mass. The true stress calculated by eq 16 is transformed into nominal (or engineering) stress by $\sigma_{\text{nom}} = \sigma_T \cdot S_i / S_0$ in which S_i and S_0 are the instantaneous and initial cross-sectional area.

MD simulations can be performed by keeping the directions perpendicular to the strained one fixed at the initial value (denoted constrained simulations, CS),⁷⁷ or in which the directions perpendicular to the strained one are allowed to relax anisotropically (denoted unconstrained simulations, UCS).⁸²

Once the stress–strain diagrams are simulated the Young's modulus E can be extracted by calculating the slope of the stress–strain curves in the linear region. Bulk modulus B can be calculated through hydrostatic compression (0.005 kbar/ps) of the samples. This is obtained according to the thermodynamic formula $B = -V(dP/dV)_T$. The thermodynamic and fracture surface energies can also be calculated.⁸²

Autocorrelation Functions and Phonon Spectrum. An autocorrelation function indicates the extent to which the system retains a memory of its previous values (or, conversely, how long it takes the system to lose its memory). A simple example is the velocity autocorrelation coefficient whose value indicates how closely the velocity at a time t is correlated with the velocity at time 0. Some correlation functions can be averaged over all the particles in the system (as can the velocity autocorrelation function) whereas other functions are a property of the entire system (e.g., the dipole moment of the sample). The value of the velocity autocorrelation function, $C_{vv}(t)$ can be calculated by averaging over the N atoms in the simulation:

$$C_{vv}(t) = \frac{1}{N} \sum_{i=1}^N \frac{\langle \mathbf{v}_i(t_0) \cdot \mathbf{v}_i(t_0 + t) \rangle}{\langle \mathbf{v}_i(t_0) \cdot \mathbf{v}_i(t_0) \rangle} \quad (17)$$

where $\mathbf{v}_i(t_0)$ is the velocity of atom i at an initial time t_0 and $\mathbf{v}_i(t_0 + t)$ is that atom's velocity at some later time t . The angle brackets again indicate the time average.

In general, an autocorrelation function such as the velocity autocorrelation coefficient has an initial value of 1 and at long times has the value 0. The time taken to lose the correlation is often called the correlation time, or the relaxation time. If the duration of the simulation is significantly longer than the relaxation time (as it should be), then many sets of data can be extracted from the simulation to calculate the correlation function and to reduce the uncertainty in the calculation. Quantities with short relaxation time (such as vibrations) can thus be determined with greater statistical precision, while quantity with a relaxation time longer than the length of the simulation cannot be determined.

The structural and dynamic results of computer simulations must often be transformed between time and frequency domains or between normal space and reciprocal space. To be compared with experiment, a time correlation function is usually transformed to produce a spectrum. From the FT of the velocity autocorrelation function the frequency spectrum $G(\omega)$, (also called vibrational density of states VDOS) is obtained

$$G(\omega) \propto \int_0^\infty C_{vv}(t) \cos \omega t \, dt \quad (18)$$

Transport Properties. The long relaxation times associated with diffusion processes demand for long simulation times to accumulate sufficient statistical accuracy. One of the simplest observable to study the dynamics of condensed matter is the self-diffusion constant D_i for a tagged particle of type i , which can be calculated from the mean-squared displacement via the Einstein relation or the velocity autocorrelation function via the Green–Kubo integrals.⁶⁰ Diffusion coefficients are usually found to vary considerably with temperature. This variation can often be expressed in terms of the Arrhenius equation from which the activation energy that an ion must overcome to migrate between two sites is obtained. Insights into the mechanism of molecular motion in time-dependent phenomena in condensed systems can be gained by Van Hove correlations functions.⁸³ In fact, the van Hove self-correlation function describes the probability density of finding an ion at a position $r(t)$ after a time t given its initial position $r(0)$ at time $t = 0$; it is thus closely related to the diffusion of the same ion and it is able to discriminate diffusion governed by a hopping process from that due to cooperative effects of ions or molecules. The distinct correlation function describes how the radial distribution function changes in response to the displacement of the ion at the center as a result of the hopping process and by the radial integration of this function it is possible to calculate the quantity of ions entering a vacancy in a given time interval.

Silica Glass

Because of its technological importance, several studies focused on silica glass have been carried out by using different interatomic potentials models.^{16,35,47,68,84}

In a very recent paper, Afyfy et al.⁸⁵ tested several rigid ionic interatomic potentials models; their results show that the best reproduction of the total static X-ray and neutron structure factors is achieved by the potential developed by Pedone et al.⁸⁴ In fact, the computed total correlation function $T(r)$, broadened using the method by Wright,⁶⁴ reported in Figure 1 was shown to fit the experimental one reported by Grimley et al.⁸⁶ with a R_x factor of 5.5% and no defect sites or oddly coordinated oxygen and silicon atoms were detected.

The calculated Si–O and O–O distances of 1.62 and 2.63 Å compared well with experimental data of 1.608 and 2.626 Å

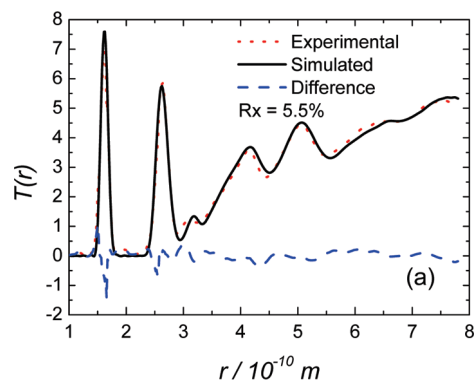


Figure 1. Computed total distribution function of SiO₂ glass, broadened using the experimental $Q_{\text{max}} = 45.2 \text{ \AA}^{-1}$, compared with the neutron diffraction data digitized from Grimley et al.⁸⁶

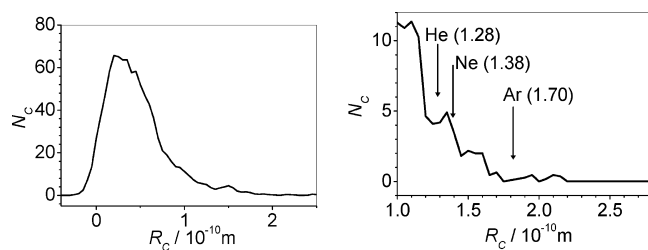


Figure 2. Distribution of nonoverlapping interstice sizes of silica glass (left). Tail of the distribution compared to the noble gas atoms dimensions (right).

as well as the O–Si–O bond angle of $109 \pm 7^\circ$ ($109.5 \pm 4^\circ$ from neutron diffraction data). Differences in the Si–O–Si bond angles were observed depending on the ensemble used. NPT simulation predicted a Si–O–Si bond angle distribution that has a full width at half-maximum (fwhm) of 37.5° and a mean angle of 147.8° while the mean value predicted by NVT simulation was 152.4° . The experimental value of mean Si–O–Si bond angle has been reported based on diffraction data by Mozzi and Warren⁸⁷ as 147.9° and recently by Neufeind and Liss as 146.7° .⁸⁸

Few works have investigated the voids size distributions in glasses despite its potentiality. In fact, in atomic systems as liquids, glasses and polymeric materials, the interstitial voids are related directly to solubility and other thermodynamic properties.^{89,90}

Malavasi et al.⁶⁸ studied the interstitial statistics in silica glass and showed that the dimensions of the interstices cover a nearly continuous range of values (see Figure 2) with a shape resembling an approximate log-normal distribution. By summing the distribution for voids larger than the radius of a specific noble gas atom the total void space accessible to that atom into the glass structure was obtained. They found 2.7 , 1.6 , and 0.29×10^{27} sites/m³ accessible to He, Ne, and Ar which compare well with the experimental ones derived by solubility data which are 2.3 , 1.3 , and 0.11×10^{27} sites/m³, respectively.⁸⁹ In particular, a very good agreement was found in the case of the concentration of interstices accessible to He and Ne atoms. These sites are created by 7-member and 8-member rings formed by corner sharing silicon oxygen tetrahedral.⁶⁸

The computed number of sites accessible to Ar atoms was systematically larger than the experimental value. The interstitial voids that can accommodate an Ar atom were surrounded by rings with more than 9-members. The comparison of ring size distribution with void size distribution calculated for different ν -SiO₂ models showed an increase in the tail of the void size

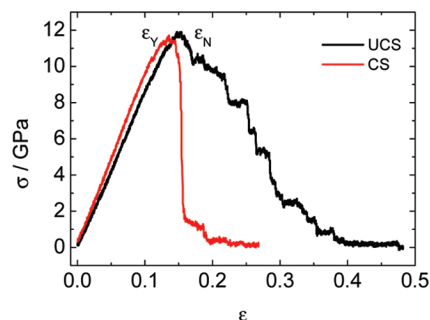


Figure 3. Stress–strain diagram of silica glass for the constrained system (red line) and the unconstrained one (black line).

distribution due to the overestimation of n -fold rings with n greater than 8 members.⁶⁸

The stress–strain behavior of crystalline and vitreous silica has been studied under different loading conditions by Simmons et al.^{77–81} and Pedone et al.⁸²

Simmons et al. applied a uniform uniaxial strain at different strain rates to glass samples of 3000 atoms where periodic boundary conditions along three directions were applied but they did not allow the system to relax along the transverse directions ($\nu = 0$, CS).

They observed that the strength of the glass increased with increasing strain rates stating that at lower strain rates the sample has more time to relieve its strain by structural rearrangement of atoms.

In a recent paper, Pedone et al.⁸² performed rigid ionic molecular dynamics simulations to study the fracture mechanism of silica glass using bigger systems (up to 12288 atoms) and allowing the system to relax along the transversal directions ($\nu \neq 0$, UCS). Figure 3 shows stress–strain relations of silica glass for the CS and UCS simulations obtained at strain rate of 10^8 s^{-1} and $T = 300 \text{ K}$.⁸²

Both Simmons et al. and Pedone et al. identified four distinct regions, (i) the elastic regime, where stress increased linearly with strain, (ii) a yield region, where stress varied very slowly with strain, (iii) a region where there was a dramatic drop in the stress with increasing strain, and finally (iv) separation of the sample corresponding to zero stress.

The CS has greater stiffness, higher intrinsic strength ($\sigma^* = 11.7 \text{ GPa}$) and lower strain at failure ($\epsilon^* = 13.4\%$) with respect to the UCS which has $\sigma^* = 10.8 \text{ GPa}$ and $\epsilon^* = 15.3\%$, which are in good agreement with experimental values at room temperature.⁹¹

Although, in both cases the fracture was observed to consist of growth and coalescence of “preexisting voids in the structure” some differences were highlighted.

While the constrained system broke catastrophically, the unconstrained one showed a recovery of the structure. Figures S.1 and S.2 of the Supporting Information show the formation and evolution of the crack for the two systems. In the constrained system (Figure S.1) a sequential breaking of the Si–O bonds occurs and results in the rapid formation of three large voids that coalesce instantaneously to break the material in just 0.2 ns.

Conversely, for the unconstrained system a gradual breaking of Si–O bonds is observed. The fragments obtained can then rearrange through the conformational and topological changes of the rings in the structure accommodating the voids that grow and coalesce until separation of the fragments in 1.4 ns of simulation and at an ultimate strain of about 40%.

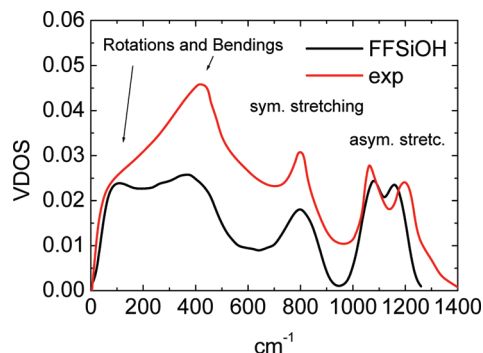


Figure 4. Vibrational density of states for the FFSiOH force field (black line) compared with the experimental data for ν -SiO₂.

The propagation of a crack in silica glass was studied by Van Brutzel et al.⁹² by using multimillion atom simulations and the potential developed by Vashishta et al.³⁵ They created a V-shaped notch on one edge of the simulation box and subjected the box to an external strain by displacing atoms in the top and bottom layers. They observed the growth and coalescence of voids and pores with radius of 50–60 Å near the crack tip.

Although the rigid interatomic potential model developed by Pedone et al.³⁶ successfully reproduced the structure and mechanical properties of silica glass, it failed to reproduce the high frequency peaks of the vibrational density of states of silica glass.⁸⁴

To overcome this problem, a new partial charge interatomic potential based on the shell model (FFSiOH) was parametrized by using periodic B3LYP simulations of silica polymorphs.³⁸

Figure 4 shows the VDOS that features a wide lower frequency band between 0 and 600 cm⁻¹ and the narrow peaks at 800 and 1100–1200 cm⁻¹ separated by a gap of 200 cm⁻¹.

The FFSiOH VDOS fits the experimental curve with an accuracy never reached before with other models potential. Taraskin et al.⁹³ simulated the VDOS of ν -SiO₂ by using the van Beest⁹⁴ and the Tsuneyuki⁹⁵ potential, which are rigid ionic models fitted on quantum mechanical and experimental observables. The deficiency of the Tsuneyuki potential results in a shift of the spectrum in the high-frequency region to lower frequencies by ~ 100 cm⁻¹ with respect to the experimental VDOS. Neither the van Beest nor the Tsuneyuki potential could reproduce a pronounced peak in the intermediate frequency region at 400 cm⁻¹ found in inelastic neutron scattering measurements.⁹⁶

Alkali Silicate Glasses

Much efforts have been devoted to the study of the effect of alkaline ions content in silica glass on the structure, transport, and elastic properties.^{20,84,97–100}

Both Pedone et al.⁸⁴ and Du et al.¹⁹ studied the effect of Li, Na, and K content on the structure of alkali silicate glasses by means of molecular dynamics simulations. Despite the different interatomic potentials used the results obtained are very similar. The local environment of the modifiers is consistent with the expectation based on their size; the smallest size that the modifier ion is, the highest the field strength is, and the best definition of its first coordination shell is reached.

Lithium shows an average coordination number of 3.0 and a distance of 1.968 Å for Li-NBO and an average Li-BO coordination number of 0.9 associated to a distance of 2.090 Å. Sodium ions are surrounded by about 5 oxygens at 2.300–2.332 Å, which compare well with EXAFS and neutron diffraction data of 2.30/2.36 Å.^{2,101,102} The K–O bond length

values range between 2.687 to 2.724 Å with a coordination number of 7.6.

The ring size distributions showed a similarity of primitive ring structure in the three types of silicate glasses with the same alkali concentration. The increase of alkali oxide concentration leads to continuous decrease of the intensity of the peak at six-membered rings and an increase of larger rings.^{19,84}

Classical molecular dynamics simulations were also employed to study the compositional dependence of the structure and the neutron static structure factors in lithium, sodium, and potassium silicate glasses.^{18,19} It was found that increasing the modifiers concentration has a significant influence on the position and shape of the first sharp diffraction peak (FSDP) in these glasses. The FSDP shifts to higher wave vectors (Q) and decreases in intensity with increasing sodium and potassium contents, while in lithium silicate glasses the FSDP shifts slightly to higher Q and its intensity changes very little with increasing lithium oxide concentration. The decrease of the intensity of the FSDP in sodium and potassium silicate glasses was found to be due to the decrease of the intermediate range order, as indicated by the decrease of characteristic repeat distance ($2\pi/Q_1$) and correlation length ($2\pi/\text{fwhm}$) that are obtained from the position and fwhm of the FSDPs, respectively. Apart from the anomaly in the intensity of the FSDP, lithium silicate glasses also show an anomaly in the change of characteristic correlation length with composition that derives from the high field strength of the lithium ions that increases the intermediate range order of the glass structure.^{18,19}

Mechanical properties such as the Young's modulus (E) and Bulk modulus (B) of the alkali silicate glass compositions were calculated by means of the second derivative method in a pioneering work by Pedone et al.⁸⁴

The overall good agreement with the experimental data obtained underlined the utility of glass simulations as a method for predicting and designing glass compositions with desired properties. Both the experimental and the calculated data showed that Li ions addition increases the Young's modulus of the silica glass, whereas a decreasing is observed when Na and K are introduced, the effect of K introduction being more marked. Different trends were obtained for the bulk modulus. Indeed, in lithium silicate glasses, B increases from 36 GPa for ν -SiO₂ to 51 GPa for 30Li₂O•70SiO₂ glass, while in sodium silicate glasses it initially decreases for low Na₂O concentrations, according to the expectation, but it overcomes the value of pure silica at 25% Na₂O concentration, with an overall increase of about 3 GPa. In the case of potassium silicate glasses, B greatly decreases for the first addition of K₂O, and then a slightly increase is observed, though the value of pure silica is never reached.

It was argued that the balance among three concurrent factors with differing consequences on the glass network is responsible of the compositional dependence of the mechanical properties. These factors are (1) depolymerization of the silica network that leads to a decrease of the elastic constants; (2) increasing of the cohesion of the glass by the establishment of alkali-NBO bonds that leads to an increase in the elastic constants as a function of the chemical nature and concentration of the modifiers; and (3) decreasing of the free volume with consequent increasing of the glass pack density leading to a minor compressibility and therefore to a greater bulk modulus.

The rigid ionic force-field derived by Pedone et al. also showed to be suited to describe the dynamics of alkaline ions in 15Li₂O•85SiO₂, 15Na₂O•85SiO₂, and 15K₂O•85SiO₂ glasses.¹⁰³

The activation energies (AE), whose values are 58.2 ± 1.3 , 65.2 ± 0.3 , and 69.7 ± 1.9 kJ/mol for Li, Na, and K ions, respectively, increase with the increasing of cation size and coordination number.

These are in good agreement with experimental data obtained by tracer diffusivity. In fact, Beier and Frischat¹⁰⁴ reported for Li in the $18\text{Li}_2\text{O} \cdot 82\text{SiO}_2$ glass a value of $\text{AE} = 65.6$ kJ/mol in the range 580–720 K, Frischat¹⁰⁵ reported for Na in the $14.8\text{Na}_2\text{O} \cdot 85.2\text{SiO}_2$ glass a value of $\text{AE} = 77$ kJ/mol in the range 518–678 K, and Evstrop'ev and Pavloskii¹⁰⁶ reported for K in the $15\text{K}_2\text{O} \cdot 85\text{SiO}_2$ glass a value of $\text{AE} = 80.5$ kJ/mol in the range 600–800 K.

Previous molecular dynamics simulations studies have revealed that ions' diffusion at low temperature occurs as an activated hopping through channels within the silica matrix, coupled to a much slower reorganization of the glass network itself.^{22,107} Alkaline ions move inside the percolation channels created by NBOs that are linked to network former ions and waggle to promote the alkaline jump.^{20,97,99,100,108}

The importance of the coupling of alkali ions with NBOs was highlighted by Heuer et al.¹⁰⁹ who showed that in lithium silicate glasses the lithium dynamics dramatically slows down leading to an increase of the activation energy by a factor of 1.7 when oxygen particles were immobilized.

Another important ingredient for the ion dynamics through the glass matrix concerns the availability of a vacant site in the neighborhood of an ion. Lammert et al.¹¹⁰ and Vogel et al.¹¹¹ showed that in lithium silicate systems the number density of sites is only few percent than the one of ions. Thus, for most ions there is no adjacent vacant site to which they might hop; therefore, they must wait for an adjacent ion to perform a hop leaving a vacant site before they can perform a hop on their own. Thus, some cooperative hopping mechanism is required. Such correlated subsequent hops were observed by Cormack et al.⁹⁷ simulating a sodium silicate system.

When more modifiers are present in the glass, the mixed alkali effect occurs,¹¹² and even small concentrations of a second alkali species effectively slow down the total ion dynamics.

Lammert et al. suggested a description of dynamics in terms of stable sites mostly specific to one ionic species and the rate slow down was related to the limited accessibility of foreign sites.²³

Soda Lime Silicate Glasses

The soda–lime–silica system is the basic constituent of a wide range of commercially important glass products, yet there have been relatively few structural studies of glasses in this system.

Molecular dynamics simulations were used by Du and Cormack¹¹³ to study the local coordination of the sodium and calcium cations, and how the replacement of sodium with calcium changes the medium range order.

It was found that the coordination number for Ca is about six with a more ordered overall first coordination shell than for sodium. NBO are preferentially bonded to calcium, which is seen to play a similar modifying role as sodium in the structure (Figure 5). On substitution, Ca replaces sodium, entering the modifier-rich regions in the structure and the ring size distribution did not change upon the substitution.

Pedone et al.¹⁰³ compared the calculated elastic properties of the $25\text{Na}_2\text{O} \cdot 75\text{SiO}_2$ and $15\text{Na}_2\text{O} \cdot 10\text{CaO} \cdot 75\text{SiO}_2$ glasses showing that the Young's modulus increases when sodium ions are substituted by calcium ones. This is a consequence of the stronger Ca–O bond strength with respect to the Na–O one.

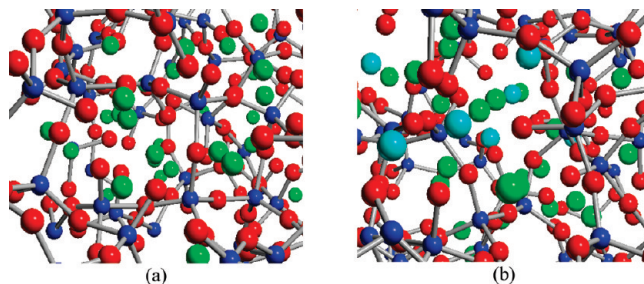


Figure 5. (a) Snapshot of channels created by Na ions (light green) in the $25\text{Na}_2\text{O} \cdot 75\text{SiO}_2$ glass and (b) channels created by Ca (light cyan) and Na (light green) in the $15\text{Na}_2\text{O} \cdot 10\text{CaO} \cdot 75\text{SiO}_2$ glass.

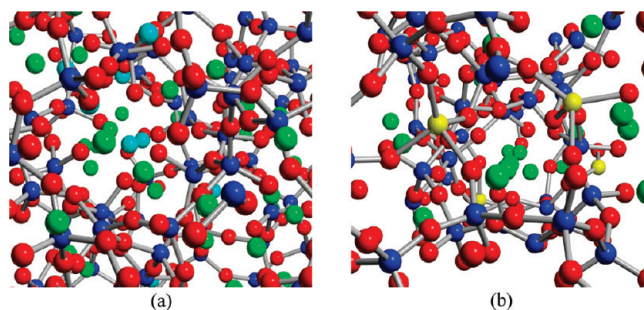


Figure 6. (a) Snapshot of channels created by Ca (light cyan) and Na (light green) in the $15\text{Na}_2\text{O} \cdot 10\text{CaO} \cdot 75\text{SiO}_2$ glass. (b) Snapshot of a percolation channel in $15\text{Na}_2\text{O} \cdot 10\text{MgO} \cdot 75\text{SiO}_2$ glass. Mg ions (light yellow) enter into the Si–O (Si light blue and O red) network acting as network former. Na ions (light green) are free to diffuse hopping among related sites. Ions have been visualized with equal radii.

Moreover, Ca ions lead to a more packed structure and increases the cohesion energy of the glass.

In the same work, it was shown that the AE of the Na ions increases by increasing the ratio $\text{CaO}/\text{Na}_2\text{O}$ in agreement with previous studies.²⁵ In fact, the activation energy of Na diffusion increases from 54.5 ± 1.3 kJ/mol for the $25\text{Na}_2\text{O} \cdot 75\text{SiO}_2$ glass to 71.4 ± 0.8 kJ/mol for $15\text{Na}_2\text{O} \cdot 10\text{CaO} \cdot 75\text{SiO}_2$ glass. Upon substitution, Ca ions lie in the percolation channels created by ion modifiers (Na and Ca) but, having diffusivities several order of magnitude smaller, obstruct the channels and hamper Na diffusion.

Recently, the effect of the replacement of CaO for MgO on the structural, transport and elastic properties of soda lime glasses with composition $15\text{Na}_2\text{O} \cdot (10 - x)\text{CaO} \cdot x\text{MgO} \cdot 75\text{SiO}_2$ with $x = 0, 5$, and 10 mol has also been studied.¹¹⁴ Magnesium was found to be 4-fold coordinated and to participate in the network, forming Si–O–Mg chains of tetrahedra (see Figure 6b), therefore acting as a pseudonetwork former.

The diffusivity of Na ions was found to increase by increasing the MgO/CaO ratio;¹¹⁴ in fact, the activation energy decreases from 71.3 to 65.8 kJ/mol for $15\text{Na}_2\text{O} \cdot 10\text{CaO} \cdot 75\text{SiO}_2$ and $15\text{Na}_2\text{O} \cdot 10\text{MgO} \cdot 75\text{SiO}_2$ glasses, respectively. The explanation was attributed to the different behavior of Ca and Mg. Mg, acting as a network former (see Figure 6b), does not obstruct the percolation channels, therefore Na ions are free to diffuse in the structure hopping among preferential sites inside channels with a mechanism previously described.^{20,97–100}

The computed elastic properties E , G , B , and ν were successfully compared to experimental data and were seen to decrease in magnitude on replacing CaO with MgO.

The simulations of the vibrational density of states showed equal force constants for the Mg–O and Ca–O bonds, thus attributing the chemical strength of the cations in the glass to their coordination number.¹¹⁴ The coordination of atoms plays

an important role in the elastic behavior of materials because when atoms are highly coordinated the deformations of M–O–M and O–M–O angles are hampered by steric effects.

Rare-Earth Silicate Glasses

Glasses doped with trivalent rare-earth ions RE^{3+} are used in telecommunication systems as optical amplifiers. Since the cross section of the stimulated emission process of RE^{3+} ions, responsible for the optical amplification effect, is directly proportional to the content of the RE^{3+} ions, large concentrations of the emitter ions have to be homogeneously distributed in the host glass. Unfortunately, RE^{3+} ions in glasses tend to form clusters at short distances of few angstroms and/or interact over large distances of about 20–100 Å. This clustering effect produces a quenching of luminescence intensity by means of ion–ion interaction mechanisms, thus the enhancing of the solubility of RE^{3+} ions in SiO_2 glass by codoping with certain oxides has been widely explored.¹¹⁵

From a structural point of view, the clustering of RE^{3+} ions in the glass can be considered as the number of other RE^{3+} ions linked via oxygen to a given RE^{3+} ion, this corresponds, in average, to the coordination number of the RE^{3+} – RE^{3+} partial correlation function.¹¹⁶ Unfortunately, experimental structural studies on RE^{3+} -doped glasses hardly provide such quantitative information which are instead easy to get from molecular dynamics simulations.

The issue of clustering in La_2O_3 – Na_2O – SiO_2 glasses was first addressed by Park and Corrales¹¹⁷ who showed that La dimers, trimers connected by La–O–La bonds were inhomogeneously distributed in the simulation box. The clustering was found to be driven by network relaxation and by the energetic of forming La–O–La linkages.¹¹⁸ The latter was calculated by determining the mean force and the potential of mean force of two La ions approaching one another. The host glass network was treated as a solvent that accommodates the solute ions by forming a Si-free cavity about them. As the cations were brought closer together, the network responded by pushing out the Si ions and maintaining the O ion population so as to increase the screening of the repulsive forces between the solute ions with net effect the cavity coalescence.¹¹⁹

The microheterogeneous cluster structures of La ions were found to be different with respect to the Na channels in silicates. The difference likely due to the high valence of La cation that leads to a tightly bound first nearest-neighbor shell.

The structure of erbium doped sodium silicate glasses $1\text{Er}_2\text{O}_3$ – $x\text{Na}_2\text{O}$ – $(99-x)\text{SiO}_2$ ($x = 0, 10, 20, 30$) were studied by Du and Cormack.⁴¹ They found that the Er–O bond length increases from 2.19 Å in silica to 2.26 in sodium silicate glass and the erbium coordination numbers also increased from 4.9 to 5.8, which is in perfect agreement with EXAFS. The bond length increasing is due to the rigid nature of the tetrahedral SiO_2 network. In fact, the main effect of doping SiO_2 glass with Er^{3+} oxide is to introduce nonbridging oxygen NBO atoms. Er^{3+} ions are usually bonded to these NBOs for charge balance. However, since a single Er^{3+} ion is charge balanced by three NBOs, the available number of NBOs becomes insufficient to satisfy the coordination of homogeneously distributed dopant ions. Therefore, Er^{3+} ions tend to form clusters to share their oxygen coordination. The ring size distribution and Q_n distribution results showed that the silicon–oxygen network is gradually depolymerized with the addition of Na_2O to silica glass. Erbium environment is more symmetric and more uniform in sodium silicate glasses than in vitreous silica because of the more open structure of sodium silicate glasses which makes erbium ions

easier to adopt favorable coordination environments and clustering is less evident.

In a very recent work by Afify and Mountjoy,⁸⁵ the force field derived by Pedone et. al.³⁶ was extended to include the Eu^{3+} –O interactions and subsequently used to study the clustering of Eu^{3+} ions incorporated in glasses with composition $x\text{Eu}_2\text{O}_3$ – $(100-x)\text{SiO}_2$ ($x = 0, 1, 2, 3, 4$, and 5 mol %). They found that at 1 mol % Eu_2O_3 most of Eu^{3+} ions are isolated in silica host but have a strained coordination with typically three short bonds to NBOs plus one long bond to BO. The Eu^{3+} –O coordination number increases from 4.5 to 4.9 atoms due to an increase in sharing of NBOs. This follows an increase in the Eu^{3+} – Eu^{3+} coordination number due to an increasing clustering of Eu^{3+} ions. Eu^{3+} clustering is already present in the sample with 1 mol % Eu_2O_3 , but most of the clustered Eu^{3+} ions are present only in pairs. As Eu_2O_3 content increases in the glass, both the proportion of Eu^{3+} ions involved in clusters and the size of individual clusters increase, and signs of phase separation become apparent for 5 mol % Eu_2O_3 .

Thanks to the high statistical quality of the calculations the authors argued that the very broad Eu^{3+} – Eu^{3+} partial correlation function, that reflect a very disordered nature of this coordination shell, was the main reason why it is not detectable by EXAFS measurements.¹¹⁶

Phospho-Silicate Bioactive Glasses

In the early 1970s, Hench and co-workers¹²⁰ showed that melt-derived silicate glasses with composition 45% SiO_2 , 24.5% CaO , 24.5% Na_2O , and 6% P_2O_5 (named 45S5 hereafter) were bioactive. Glasses of this type show a strong active response after they are implanted in the human body or put in contact with acellular simulated body fluid (SBF).¹²¹

A series of chemical transformations occurs after implantation, which leads to the growth of a layer of crystalline hydroxyl-carbonate apatite (HCA) on their surface. This layer then interacts with and incorporates biomolecules such as collagen, whereas further cellular steps lead to a strong and stable chemical bond between the glass and human hard (bone) and, in some cases, soft (muscles) tissues. Therefore, bioglasses have been widely used in restorative and regenerative medical applications.¹²² These applications are generally based on the 45S5 composition because of its ability to induce a formation of new bone very shortly after implant.

By altering the glass composition, one can in principle tune the rate of HCA formation and the glass physical properties for a specific application; however, a more rational control of the bioactivity could be achieved by establishing the microscopic, structural basis for the bioactive behavior and its relation with the composition.

The microscopic structure of these materials have been explored in higher detail by several classical molecular dynamics simulations employing both the rigid ionic model^{59,123} and the shell-model.^{46,124,125} The coordination environment of network formers and modifiers,^{59,126} the tendency to form clusters and inhomogeneities,^{46,127} the occurrence of chain and ring nanostructures¹²⁵ and the effect of phosphorus inclusion were discussed in relation to bioactivity,¹²⁴ and medium-range order.¹²³ All these computational techniques showed to be in perfect agreement in reproducing short-range geometrical features such as the coordination and bond distances of network formers and modifiers.⁵⁹ On the contrary, small differences were obtained in the structural parameters commonly used to describe the medium range order in amorphous glasses such as Si–O–Si and Si–O–P angles and Q_n distributions. In fact, molecular

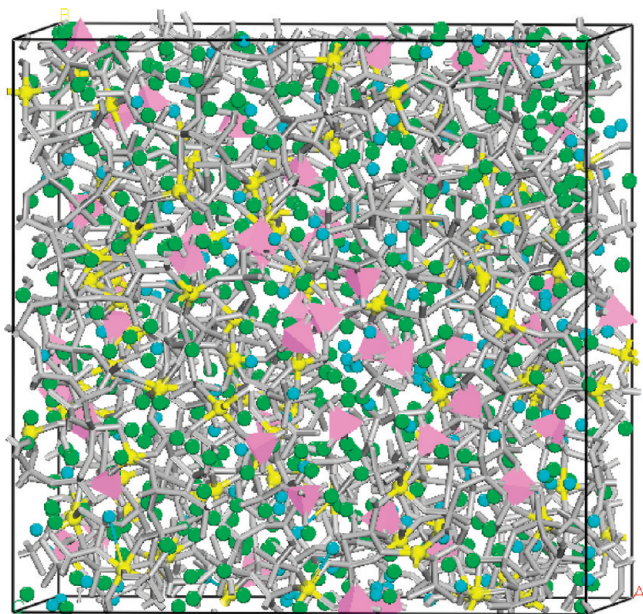


Figure 7. Snapshots of the three-dimensional structure of $46.2\text{SiO}_2 \cdot 24.3\text{Na}_2\text{O} \cdot 16.9\text{CaO} \cdot 2.6\text{P}_2\text{O}_5 \cdot 10\text{MgO}$ glass. Violet tetrahedra represent the PO_4 units, yellow spheres represent Mg, green spheres represent Na, cyan spheres represent Ca, while gray sticks represent the Si–O framework.

dynamics simulations by using the rigid ionic model overestimates the Si–O–Si and Si–O–P intertetrahedral angles with respect to molecular dynamics simulations performed with the shell-model, static QM methods and Carr–Parrinello molecular dynamics. This leads to worse Q_n distributions compared to experimental data. However, the trends in the distribution is captured by the rigid ionic model.

Among the computational studies found in literature, few works have been devoted to the simulation of the effect of doping ions on the structure of bioactive glasses^{42,128} and despite the paramount importance of transport, elastic and mechanical properties no such simulations have been carried out so far.

A first attempt was carried out by Pedone et al.¹²⁹ who studied the effect of the replacement of CaO for MgO on the structural properties of the 45S5 Bioglass. The results confirmed the complexity of the local environment of Mg ions, which are coordinated predominantly by 5 nonbridging oxygens of different TO_4 tetrahedra ($T = \text{Si/P}$) leading to large rings in the structures. A rough correlation between the average dimension of the rings found in the structure and the computed Young's modulus was obtained. The Young's modulus decrease at low Mg-content reaching a minimum for the $46.2\text{SiO}_2 \cdot 24.3\text{Na}_2\text{O} \cdot 16.9\text{CaO} \cdot 2.6\text{P}_2\text{O}_5 \cdot 10\text{MgO}$ glass. A snapshot of the MD derived glass structure of this composition is reported in Figure 7. Mg was found to be homogeneously distributed in the silica rich region together with Ca and Na ions but was almost totally absent from the Ca–Na-phosphate rich regions.

Lusvardi et al.^{42,128} performed molecular dynamics simulation study of the nanosegregation promoted by the incorporation of fluorine ions in bioactive glasses. The results showed that the high affinity of fluorine for Na and Ca modifier cations determines the separation of a highly polymerized phospho-silicate matrix from an ionic phase rich in Na, Ca, and F (figure 8). Conversely, fluorine-free bioglasses favor the separation of a phosphate-rich phase from a silica-rich one.^{46,127}

The aforementioned studies as well as the relationships between the microscopic structure and bioactivity of phospho-

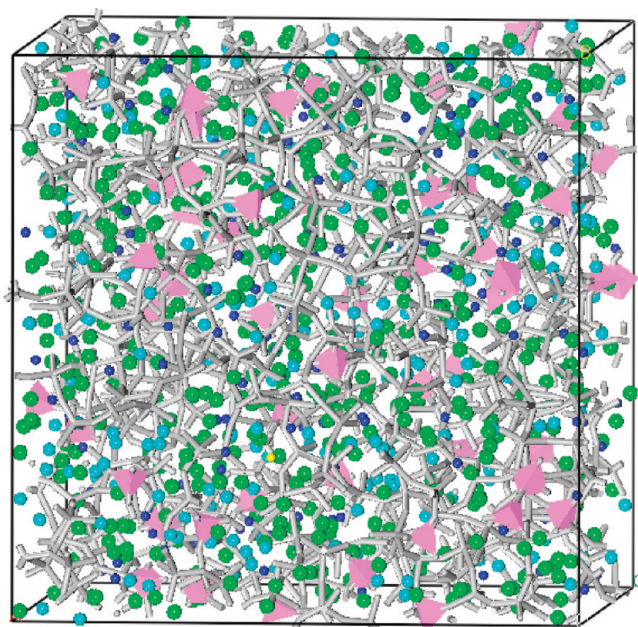


Figure 8. Snapshots of the $46.2\text{SiO}_2 \cdot 24.3\text{Na}_2\text{O} \cdot 16.9\text{CaO} \cdot 2.6\text{P}_2\text{O}_5 \cdot 10\text{CaF}_2$ glass. Violet tetrahedra represent the PO_4 units, blue spheres represent F, green spheres represent Na, cyan spheres represent Ca, while gray sticks represent the Si–O framework.

silicate glasses have been very recently reviewed by Tilocca,¹³⁰ therefore no further detail are given here.

Summary and Concluding Remarks

Results of classical molecular dynamics and structural property calculations of silica based glasses have been presented, along with the basic methodology for performing such simulations. These have been shown to reproduce many important features of silica, alkaline-silicate, soda-lime, rare-earth silicate, and phospho-silicate glasses.

The use of two-body interatomic potentials with partial charges in the framework of the rigid ionic model has been shown to reproduce well the short-range order, mechanical, and transport properties whereas the addition of polarizability terms are needed to obtain a better description of the medium range order and vibrational properties of such systems.

Reactive force-fields,^{131,132} which enable accurate dynamic simulations of bond breaking processes in large systems, have been recently developed for silica but they need to be tested and extended for multicomponent glasses. This would open up new perspectives to the study of reactivity at the glass surfaces and fracture mechanisms in a wet environment.

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Supporting Information Available: Tables of potentials parameters used together with geometric parameters of oxides, hydroxydes, silicate and aluminosilicate crystal optimized are available as well as figures of fracture mechanism in silica glass. This material is available free of charge via the Internet at <http://pubs.acs.org>.

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